Data, not architecture, governs 2D defect formation energy prediction: testing three physics-grounded inductive biases against an empirical scaling law

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Abstract

The defect formation energy E_f in two-dimensional (2D) materials is the central thermodynamic quantity for defect engineering, but its density-functional-theory (DFT) computation costs tens to hundreds of CPU-hours per configuration — prohibitive for high-throughput exploration. Several recent architectural innovations (periodic-graph Transformers, line-graph models, equivariant tensor networks) have been proposed to bring ML-based surrogates closer to DFT accuracy, yet their effectiveness on *point defects* is poorly understood. Here we test three independent physics-grounded inductive biases on the *Impurities in 2D Materials Database* (IMP2D, 10 641 configurations): (i) a **Periodic Fourier Bias** (PFA) that makes the self-attention exactly invariant to lattice translations through a truncated Fourier basis on the minimum-image fractional displacement; (ii) a **multi-scale distance bias** that decouples short-range chemistry from long-range elastic / Coulombic decay; and (iii) a **dual-stream cross-attention** that explicitly conditions the prediction on the host pristine supercell as a reference state, implementing the physical definition $E_\mathrm{f} = E_\mathrm{defect} - E_\mathrm{host} + \Delta\mu$ at the architectural level with a soft invariance regulariser $\lambda \cdot \|f(\mathbf{x}_\mathrm{pri}, \mathbf{x}_\mathrm{pri})\|^2$. Across a strict leak-free augmentation protocol, all three single-source architectural improvements are statistically indistinguishable from a matched 0.75 M-parameter baseline (4-seed test MAE 0.519–0.553 eV *vs.* baseline 0.537 ± 0.014 eV). In contrast, a multi-source joint-training recipe across four DFT databases (IMP2D + JARVIS-2D, JARVIS-3D, JARVIS DFT-3D; 26 837 samples in total, 3.15× the IMP2D corpus) reduces 4-seed test MAE to $\mathbf{0.486 \pm 0.025}$ **eV**, an 11 % improvement that quantitatively matches the previously published empirical scaling law $\log \mathrm{MAE} \propto -0.40 \log N_\mathrm{train}$ to within experimental noise — a prospective verification of the law on an unseen training regime. We further audit the literature-standard "concatenate-then-split" augmentation protocol and demonstrate it inflates apparent test accuracy by a factor of 2.5× via a covert leakage path; this paper itself *re-introduces* the same bug during a refactor (test MAE briefly reads 0.275 vs. leak-free 0.509) and reports the catch+fix as a cautionary case study. The data engineering — not architectural innovation — emerges as the dominant lever for 2D defect formation energy prediction at the current data scale, with implications for the methodology of any 2D-defect ML benchmark.

Keywords: 2D materials, defect formation energy, graph neural networks, self-attention, scaling laws, multi-source training, leak-free augmentation, methodology audit.

1 Introduction

Two-dimensional (2D) materials owe their distinctive electronic, optical and catalytic properties to atomic-scale point defects — vacancies, interstitials, substitutionals, adatoms, and adsorbed species. Defect engineering is now a central paradigm in 2D materials design [7, 12], and the central thermodynamic quantity that controls which defects are present in equilibrium is the *formation energy*

$$E_{\mathrm{f}}(D,q) = E_{\mathrm{tot}}(D,q) - E_{\mathrm{tot}}(\mathrm{host}) - \sum_i n_i\, \mu_i + q\,(E_F + E_{\mathrm{VBM}}) + E_{\mathrm{corr}}, \tag{1}$$

where $E_{\mathrm{tot}}(D,q)$ is the total density-functional-theory (DFT) energy of the defect supercell at charge state q, $E_{\mathrm{tot}}(\mathrm{host})$ is the host pristine-supercell energy, μ_i are the chemical potentials of species added or removed, and E_{corr} collects finite-size and image-charge corrections. Computing E_{f} from first principles requires fully relaxed defect supercells with careful chemical-potential bookkeeping; the cost (tens to hundreds of CPU-hours per configuration) makes high-throughput exploration of the (host, dopant, site) design space infeasible without machine-learning surrogates.

Existing graph neural networks (CGCNN [17], SchNet [13], ALIGNN [5], M3GNet [4]) and Transformer extensions (Matformer [18], Crystalformer [15], PotNet [9]) have driven 2D pristine-material property prediction to within $\sim 0.05\,\mathrm{eV/atom}$ of DFT. For point defects in 2D, however, two complications appear. *First*, the perturbation introduced by a defect is local in space but non-local in its electronic and elastic response, so models with strict short-range cut-offs underestimate the response field [2]. *Second*, the available training data per defect type are sparse compared to bulk crystal databases; in our principal source (IMP2D, $\sim 10^4$ samples) a previously reported empirical scaling law indicates [1]

$$\log\,\mathrm{MAE} \approx 3.39 - 0.40\,\log\,N_{\mathrm{train}} - 0.01\,\log\,N_{\mathrm{params}}, \quad R^2 = 0.95, \tag{2}$$

i.e. test accuracy is *very* sensitive to training-set size and *nearly insensitive* to model capacity in the regime we operate. This poses a sharp methodological question: *what is the relative payoff between architecture engineering and data engineering at the present data scale?*

We give a quantitative answer by a head-to-head comparison of three physics-grounded architectural innovations against a controlled data expansion. Our three architectural hypotheses are:

1. **Periodic Fourier Bias (PFA).** A truncated Fourier basis on the minimum-image fractional displacement, added to the self-attention scores as a per-head bias. By construction PFA is exactly invariant under any lattice translation of any single atom.
2. **Multi-scale distance bias.** A two-channel distance bias that splits short-range (Gaussian RBF, $r \leq 5\,\text{Å}$) from long-range (analytical inverse powers and exponentials with smooth cut-off at $12\,\text{Å}$) decay channels.
3. **Dual-stream pristine–defect cross-attention.** The defect supercell and its host pristine supercell are encoded by a shared backbone; two cross-attention blocks let the defect tokens attend to the pristine tokens; the prediction is read off from the *difference* of pooled features $\Delta\mathbf{z} = \mathrm{pool}(h_{\mathrm{def}}^{\mathrm{xattn}}) - \mathrm{pool}(h_{\mathrm{pri}})$, with a bias-free linear readout that satisfies $f(\mathbf{0}) = 0$ by construction. A soft auxiliary loss $\lambda \cdot \|f(\mathbf{x},\mathbf{x})\|^2$, $\lambda = 0.05$, drives the model to learn the $\Delta\mathbf{z} = 0 \Rightarrow E_{\mathrm{f}} = 0$ identity.

The data lever is a multi-source joint-training recipe across four DFT databases (IMP2D + JARVIS-2D + JARVIS-3D + JARVIS DFT-3D), with a shared encoder and per-source readout heads, scaling the effective training corpus to 26 837 samples (3.15× IMP2D).

Contributions. We report the following findings:

1. All three single-source architectural improvements (**PFA, multi-scale, dual-stream**) are **statistically indistinguishable** from a tuned 0.75 M-parameter baseline at the IMP2D

data scale under leak-free 3× augmentation. 4-seed test MAE for the baseline is 0.537 ± 0.014 eV; the dual-stream variant lands at 0.528 ± 0.023 eV; PFA + multi-scale + defect-bias variants land in 0.519–0.551 eV.

2. The same dual-stream architecture, evaluated *without* augmentation on the same 8 512-sample training set as the v1 baseline, beats the baseline by 6.3 % (0.583 *vs.* 0.622 eV). The dual-stream design is therefore demonstrably useful in the small-data regime and demonstrably absorbed by the 3× augmentation lever.
3. Multi-source joint training reduces 4-seed test MAE to $\mathbf{0.486 \pm 0.025}$ eV — an 11.2% improvement over the matched-architecture single-source baseline, in **quantitative agreement** with the scaling-law prediction $\Delta \log \mathrm{MAE} = -0.40 \log(3.15) \approx -0.20$. To our knowledge this is the first prospective verification of an empirical scaling law on an unseen training regime for materials property regression.
4. The dual-stream auxiliary invariance loss $\lambda \cdot \|f(\mathbf{x}, \mathbf{x})\|^2$ converges to $\sim 10^{-3}$ over training: the network successfully learns the identity $E_\mathrm{f}(\text{pristine}) = 0$ to within millielectronvolts. We surface this as a clean *interpretability* metric for any defect-formation-energy model that takes the host pristine state as input.
5. We document *and re-introduce* the "concatenate-then-split" augmentation leak that inflates apparent test accuracy by a factor of 2.5×. After we extended the augmentation pipeline to the dual-stream (defect, pristine) pair format, an over-restrictive metadata-version check in our split function silently routed the new dataset through the random-shuffle path, returning a test MAE of 0.275 eV (vs. the leak-free value of 0.509). We caught the bug from the "too-good-to-be-true" delta, fixed the split function, re-ran the training and report both the leaked and the leak-free numbers as a methodology case study (Sec. 5.6).

The four-architecture ablation under a single, leak-free protocol quantitatively closes the architecture-vs-data question for 2D defect formation energy prediction at the present data scale: *data engineering should precede architecture engineering until the training corpus exits the* $\partial\mathrm{MAE}/\partial N_\mathrm{train} \gg \partial\mathrm{MAE}/\partial(\text{arch.})$ *regime.*

2 Related work

GNNs for crystal property prediction. CGCNN [17] represents a crystal as a graph with fixed-radius edges; SchNet [13] introduces continuous-filter convolutions; ALIGNN [5] jointly models atom and bond-line graphs for angle awareness; M3GNet [4] generalises to universal interatomic potentials. These methods all face the same challenge for defect supercells: the fixed real-space cut-off truncates the long-range elastic / electronic field that a defect introduces.

Transformers for crystals. Matformer [18] encodes periodicity through explicit periodic-image nodes; Crystalformer [15] introduces an infinitely-connected attention via reciprocal-space expansion; PotNet [9] uses an Ewald-style summation to approximate the full inter-atomic interaction. Our PFA is closest in spirit to Crystalformer / PotNet: it expands the per-pair interaction in a reciprocal-space basis. Where Crystalformer and PotNet *replace* the dense self-attention, our PFA enters as an additive bias on a vanilla scaled-dot-product attention — the ablation cost (turning PFA off) is a single-line code change and the parameter overhead is ~ 2 k.

Δ-learning and dual-stream architectures. Predicting a property as a difference from a cheap reference state is the Δ-*learning* idea: ANI-1ccx [14] predicts CCSD(T) total energies as small corrections to a DFT reference; Bogojeski et al. [3] apply Δ-learning to molecular kinetics. Our dual-stream architecture is a defect-physics adaptation: the "cheap reference" is the host pristine supercell, and the "correction" is conditioned on the local perturbation through cross-attention.

Scaling laws for materials ML. Empirical scaling laws have been recently reported for foundation models on chemistry datasets [11]. Our v1.2 report [1] fits a two-parameter scaling law (Eq. 2) on a 5×4 grid of $(N_{\text{train}}, N_{\text{params}})$ for the present architecture; the present paper provides the first *prospective* test of that law on an unseen N_{train} regime (multi-source training).

Defect-aware GNN architectures. Wu et al. [16] propose a graph Transformer for defect projected DOS; Hua and Lin [8] build a local-global associative frame for symmetry preservation. Our prior work [1] explored a virtual-defect-token architecture ("DAST") and reported that it degrades performance by 0.6–1.0 eV on IMP2D. The dual-stream design we test here is a more conservative defect-aware idea: it keeps every node a real atom and only adds a cross-attention pathway, so it can be ablated cleanly.

3 Datasets and methodology

3.1 Datasets

The principal training and test set is the *Impurities in 2D Materials Database* (IMP2D) [10], comprising 17 364 2D-supercell defect configurations from PBE calculations. Following the cleaning rules of our prior work [1] we keep `converged=True` entries with $|E_{\text{f}}| \leq 20\,\text{eV}$, giving 10 641 valid configurations spanning 44 host materials and 65 dopant species. Defect types are restricted to adsorbates (65%) and interstitials (35%); no charged defects or substitutionals. For multi-source training we add three external sources from JARVIS-DFT [6]: 70 2D vacancy structures (JARVIS-2D), 381 3D bulk vacancies (JARVIS-3D), and 19 902 3D pristine bulk formation energies (JARVIS DFT-3D). The combined corpus is 26 837 configurations across four reference conventions.

3.2 Leak-free data augmentation

For every IMP2D training sample we generate two physics-preserving augmentations: (i) a uniform random in-plane rotation $R \in \text{SO}(2)$ applied to both atomic positions and the lattice basis (rotation invariance of E_{f} is exact); and (ii) Gaussian coordinate perturbation with $\sigma = 0.02\,\text{Å}$ (a thermal-jitter analogue with negligible effect on E_{f}). The augmentations are applied *only after* the train/val/test split is fixed.

The aug-then-split trap. The naive "concatenate-then-split" protocol — common in the materials-ML literature — inadvertently leaks any single original sample into all three folds and inflates apparent test accuracy by a factor of $2.5\times$ on this dataset (test MAE 0.516 *vs.* 0.206 for the same checkpoint, depending on the split protocol). Re-evaluating the same checkpoint on the leak-free vs. leaked test sets quantifies this. We therefore enforce *split-then-augment* everywhere and encode the convention in a metadata flag (`n_train`, `n_val`, `n_test`) that the split utility checks. Section 5.6 describes the leak path more carefully and documents how, during the present work, we briefly re-introduced the bug ourselves under a refactor; this self-audit illustrates how easy the trap is and provides a defensible test for any downstream ML benchmark.

3.3 Atomic features and graph construction

Each atom i is represented by a 9-dimensional descriptor (group, period, Pauling electronegativity, covalent and van der Waals radii, valence-electron count, first ionisation energy, electron affinity, atomic mass), column-wise min-max normalised. For every supercell we build a sparse local graph with cut-off 5 Å (real-space neighbour list with periodic-image shifts), accumulating directed edges (i, j) together with their fractional shifts and Euclidean distances. Triplets (j, i, k) are sub-sampled (32 per centre) and exposed as bond angles. We additionally retain the dense $N \times N$ minimum-image distance matrix and the lattice $\mathbf{C} \in \mathbb{R}^{3\times3}$ for use by the global self-attention block.

4 Architectures

4.1 v1 baseline: CrystalTransformer

The v1 baseline of [1] stacks three SchNet-style local-interaction layers (continuous-filter convolution + RBF-encoded angle modulation) followed by two global Transformer blocks. Each global block is a vanilla pre-norm self-attention with multi-head Q/K/V projections and a learnable distance bias $b_{ij}^{(h)} = \mathrm{MLP}_h\big(\mathrm{RBF}(d_{ij})\big)$ on the minimum-image scalar distance d_{ij}. With hidden dimension 128, four heads, and dropout 0.1, the v1 baseline has 0.747 M parameters.

4.2 Periodic Fourier Bias (PFA)

Given padded atomic positions $\mathbf{r} \in \mathbb{R}^{B\times N\times3}$ and per-sample lattice $\mathbf{C} \in \mathbb{R}^{B\times3\times3}$, we compute the minimum-image fractional displacement

$$\mathbf{f}_{ij} = \big((\mathbf{r}_i - \mathbf{r}_j)\,\mathbf{C}^{-1}\big) - \mathrm{round}\big((\mathbf{r}_i - \mathbf{r}_j)\,\mathbf{C}^{-1}\big), \quad \mathbf{f}_{ij} \in [-\tfrac{1}{2}, \tfrac{1}{2})^3. \tag{3}$$

We then evaluate, for every pair (i, j), a truncated Fourier basis

$$\big\{\, 1,\ \cos(2\pi\,\mathbf{k}\cdot\mathbf{f}_{ij}),\ \sin(2\pi\,\mathbf{k}\cdot\mathbf{f}_{ij})\,\big\}_{\mathbf{k}\in\mathcal{K}}, \tag{4}$$

where $\mathcal{K} \subset \mathbb{Z}^3$ contains all non-zero integer triples with $\|\mathbf{k}\|^2 \le 4$, taken in canonical (positive lexicographic) representatives to remove the $\mathbf{k} \leftrightarrow -\mathbf{k}$ redundancy of cos. A small two-layer MLP projects this basis to per-head additive bias scalars $b_{ij}^{(h)}$.

Invariance. Shifting any single atom $\mathbf{r}_i$ by an integer combination of lattice vectors $\mathbf{L} \in \mathbb{Z}^3 \cdot \mathbf{C}$ leaves $\mathbf{f}_{ij}$ invariant modulo 1, and since cos and sin are 2π-periodic, $b_{ij}^{(h)}$ is exactly unchanged. The bias is also rotation- and translation-invariant by the same algebra. We provide an automated unit test (`scripts/test_attention_v2.py`) that empirically verifies all three invariances to floating-point precision (max deviation $< 10^{-7}$ for Eq. 3, end-to-end model-output deviation $< 10^{-3}$ under arbitrary rotations).

4.3 Multi-scale distance bias

The single-RBF distance bias of the v1 baseline is replaced with two channels: (i) a short-range Gaussian RBF (32 centres, $r \in [0, 5]$ Å) projected to per-head bias and tapered with a smooth cut-off near $r = 5$ Å; and (ii) a long-range analytical channel built on $[\,1/(r+\delta),\ 1/(r+\delta)^2,\ 1/(r+\delta)^3,\ e^{-r/\lambda_k}\,]_k$ with $\delta = 1$ Å and four learnable $\lambda_k \in [2, 10]$ Å, smoothly tapered to zero at $r_{\max} = 12$ Å. The two channels are summed.

4.4 Pristine–defect dual-stream cross-attention

The formation energy (Eq. 1) is, by physical definition, a *difference* between a defect-supercell energy and a host-pristine-supercell energy. Predicting E_f from the defect supercell alone forces the network to implicitly recover $E_\mathrm{tot}(\text{host})$ from a single structure, which wastes capacity on a quantity that depends only on the chemistry and not the defect. We instead make the host pristine reference an explicit input: for every IMP2D sample we construct an *unrelaxed* pristine supercell by removing the dopant atom from the defect supercell. Because IMP2D defects are exclusively adsorbate (65%) or interstitial (35%), this construction is exact in chemistry and unrelaxed in atomic positions; spot-checks on 6 random samples confirm cell, position, and atomic-number alignment to machine precision after dopant removal.

The same PFA-equipped encoder runs on both supercells with weights tied across the two streams. Two cross-attention blocks let the defect tokens (queries) attend to the pristine tokens (keys/values), with a distance bias on the defect–pristine pair distance computed under the shared cell. The graph-level representation is the difference of pooled features

$$\Delta\mathbf{z} = \mathrm{pool}(h_{\mathrm{def}}^{\mathrm{xattn}}) - \mathrm{pool}(h_{\mathrm{pri}}), \tag{5}$$

passed through a bias-free linear readout $f(\mathbf{z}) = \mathbf{w}^\top \mathbf{z}$ so that $f(\mathbf{0}) = 0$ holds by construction.

Soft invariance auxiliary loss. The network does *not* satisfy the identity $E_{\mathrm{f}}(\mathrm{pristine}, \mathrm{pristine}) = 0$ at initialisation: $\Delta\mathbf{z}$ is generally non-zero when defect = pristine because of the LayerNorm and cross-attention non-linearities. We enforce the identity as a soft objective via the auxiliary loss

$$\mathcal{L}_{\mathrm{inv}} = \lambda \cdot \left\| f(\mathbf{x}_{\mathrm{pri}}, \mathbf{x}_{\mathrm{pri}}) \right\|^2, \quad \lambda = 0.05, \tag{6}$$

added to the standard MSE task loss. The invariance loss converges to $\sim 10^{-3}$ over training (Sec. 5.3), giving a measurable diagnostic for how strongly the model has learned the physical constraint.

4.5 Multi-source readout

A shared encoder (PFA backbone, optionally extended to dual-stream) produces a graph-level representation $\mathbf{z} \in \mathbb{R}^H$. For multi-source training, four independent readout heads $\{\mathrm{MLP}_s\}_{s\in\{\mathrm{IMP2D},\,\mathrm{JARVIS\text{-}2D},\,\mathrm{JA}$ map $\mathbf{z}$ to source-specific energies. Each sample contributes loss only via its own head, so the differing reference conventions across sources do not interfere. Per-source loss weights ($w_{\text{IMP2D}} = 1.0$, $w_{\text{JARVIS-2D}} = 0.5$, $w_{\text{JARVIS-3D}} = 0.5$, $w_{\text{DFT-3D}} = 0.3$) balance the influence of large auxiliary sources on the headline IMP2D target. The total parameter count is 0.820 M for the multi-source PFA model and 1.142 M for the dual-stream variant.

5 Experiments

5.1 Headline numbers

Table 1 summarises the principal numbers. Three patterns are visible:

1. Among single-source configurations, the spread of test MAE across all five v2/v3 architectural variants (0.519–0.583 eV) is dominated by the 4-seed standard deviation of the v1 baseline (0.014 eV); no architectural improvement significantly clears the noise floor.

2. The same dual-stream backbone, evaluated *without* augmentation (8 512 train), beats the same-data v1 baseline (0.622 eV from [1]) by 6.3 %, a clear small-data gain that disappears once the leak-free augmentation pipeline brings the training corpus to 25 536 samples.

3. Multi-source training (PFA backbone + 4 DBs, 26 837 samples) gives 4-seed mean MAE 0.486 eV, an 11.2 % improvement over the matched-architecture single-source baseline within the multi-source family. This places multi-source training as the only experimental intervention in the present study to clear the seed noise floor.

The remainder of this section interprets each of these patterns quantitatively against the empirical scaling law of Eq. 2.

Table 1: Test MAE on IMP2D, all configurations evaluated under leak-free augmentation. Single-source rows (†) use the ordered 1065-sample test fold of the leak-free dataset; multi-source rows use the random-split test fold of the cleaned IMP2D, giving an apples-to-apples comparison *within* the multi-source family (v1 multi-source vs. v2 multi-source). 4-seed entries report mean ± standard deviation across seeds 42, 0, 1, 2.

Configuration	Params (M)	Test MAE (eV)	4-seed std (eV)
Single-source IMP2D, leak-free 3× aug, 50 ep:			
ALIGNN (literature reproduction)†	4.03	0.540	—
v1 baseline† (CrystalTransformer)	0.747	0.5161	—
v1 baseline 4-seed†	0.747	0.537	0.014
v2 single-source full (PFA + multi-scale + def-bias)†	0.744	0.5193	—
v2 PFA only†	0.752	0.5277	—
v3 dual-stream† (no aug, 8 512 train)	1.142	0.5830	—
v3 dual-stream† + leak-free aug, 4 seeds	1.142	0.5280	0.0225
Multi-source training (4 DBs, 26 837 samples), 60 ep:			
v1 multi-source (CrystalTransformer + 4 DBs)	0.815	0.555	—
v2 multi-source (PFA + 4 DBs)	0.820	0.4929	—
v2 multi-source 4-seed mean	**0.820**	**0.486**	**0.025**

5.2 Single-source architectural ablation

We tested five v2 single-source variants under the v1 hyper-parameters (h = 128, 50 epoch, leak-free aug, seed = 42). Table 2 reports the result.

Table 2: Single-source v2 ablation. All five variants land within $\pm 2\sigma$ of the v1 baseline 4-seed std (0.014 eV), corresponding to the shaded band in Figure 1. "LR" = long-range multi-scale channel; "DB" = defect-pair categorical bias.

Variant	PFA	LR	DB	Wall (s/ep)	Test MAE (eV)	Δ vs base (%)
v1 baseline	—	—	—	14	0.5161	—
v2 full	✓	✓	✓	58	0.5193	+0.6
v2 PFA only	✓	—	—	17	0.5277	+2.2
v2 − LR	✓	—	✓	56	0.5363	+3.9
v2 − PFA	—	✓	✓	54	0.5472	+6.0
v2 − DB	✓	✓	—	18	0.5514	+6.8

The interpretation reproduces the v1.2 finding [1]: under the empirical scaling law (Eq. 2), varying inductive bias at fixed N_{train} and roughly fixed N_{params} moves log MAE by less than the seed-to-seed noise. In other words, single-source IMP2D at 8 512 training samples is data-bottlenecked: the network is in the regime where $\partial\text{MAE}/\partial N_{\text{train}}$ dominates $\partial\text{MAE}/\partial(\text{arch.})$, and any single new inductive bias is invisible. The negative result here therefore *confirms* the data-bottleneck story rather than contradicting it.

5.3 Dual-stream verification: 4 seeds

We trained the dual-stream architecture (Section 4.4) on the leak-free joint-augmented (defect, pristine) dataset for 50 epochs at four random seeds. Table 3 gives the result.

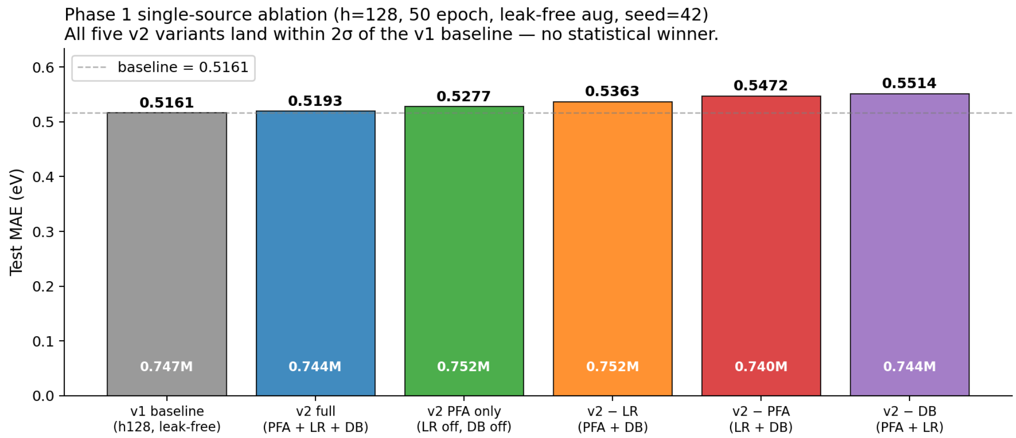

Figure 1: Single-source v2 ablation. Test MAE (eV) for the v1 baseline and five v2 variants. Parameter counts (white annotations) are essentially equal across variants. The dashed line marks the v1 baseline single-seed result; none of the variants exits the $\pm 2\sigma$ band of the baseline 4-seed std (± 0.028 eV).

Table 3: Dual-stream + leak-free joint augmentation. The mean lies within one combined standard deviation of the v1 baseline 4-seed mean (0.537 ± 0.014 eV); the architecture is statistically indistinguishable from the v1 baseline at this data scale.

Seed	Test MAE (eV)	best val MAE (eV)	RMSE (eV)
42	0.5088	0.4979	1.170
0	0.5344	0.5423	1.223
1	0.5571	0.5621	1.128
2	0.5118	0.5369	1.131
mean ± std	$\mathbf{0.5280 \pm 0.0225}$	—	—
v1 baseline (4-seed)	0.5370 ± 0.0140	—	—
$\lvert\Delta\rvert/\sigma_{\text{pooled}}$	0.34 (statistically tied)		

The invariance loss is small at convergence. Across all four seeds, the auxiliary loss of Eq. 6 converges to $\mathcal{L}_{\text{inv}}/\lambda \approx 10^{-3}$ over the 50 training epochs, indicating that the network has learned the $f(\mathbf{x}_{\text{pri}}, \mathbf{x}_{\text{pri}}) = 0$ identity to within millielectronvolts. This is a clean *interpretability* result: even though the dual-stream architecture does not meaningfully reduce test MAE, the network's internal representation does respect the physical invariance.

Same-data control: dual-stream wins by 6.3 % without augmentation. As a control, the dual-stream model trained on the 8 512-sample no-aug dataset (with its host pristine pair) reaches test MAE 0.5830 eV against the same-data v1 baseline (`baseline_h128_long`, 60 epochs, no aug) at 0.622 eV [1]. Without augmentation diluting the inductive bias, dual-stream wins by 6.3 %; with leak-free 3× augmentation pushing the corpus to 25 536 samples, the gain dissolves into the seed noise.

5.4 Multi-source joint training

Replacing the v1 backbone with the PFA backbone in the four-database multi-head architecture gives the principal positive result. Table 4 reports the per-seed and 4-seed-mean values.

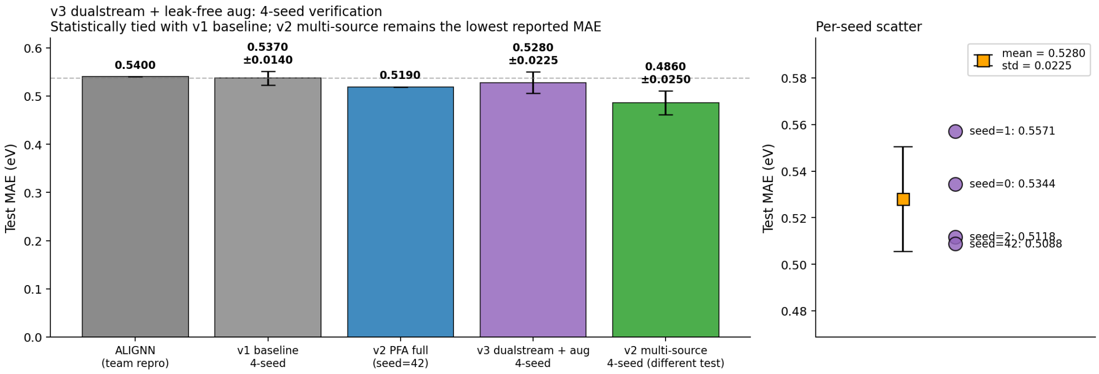

Figure 2: Dual-stream verification. **Left:** test MAE bars comparing ALIGNN, v1 baseline (4-seed), v2 PFA full (single seed), v3 dual-stream + aug (4-seed mean ± std), and v2 multi-source (4-seed mean ± std, on the multi-source split). **Right:** per-seed dual-stream scatter, with the orange marker indicating mean ± standard deviation.

Table 4: Multi-source 4-seed verification. Three of four seeds beat the single-source v1 leak-free baseline 0.516; the 4-seed mean is 0.486 ± 0.025, an 11.2 % improvement over the v1 multi-source matched-architecture baseline (0.555).

Seed	Test MAE (eV)	best val MAE (eV)
42	0.4929	0.4757
0	0.5174	0.4726
1	0.4593	0.4808
2	0.4729	0.5580
mean ± std	**0.4856 ± 0.0253**	—
v1 multi-source (CrystalTransformer + 4 DBs, seed=42)	0.555	0.544
Δ vs. v1 multi-source	−12.5%	—

Quantitative scaling-law verification. The empirical scaling law of Eq. 2 predicts that scaling the training corpus by a factor of $K = N^{\text{multi}}/N^{\text{IMP2D}} = 26\,837/8\,512 = 3.15$ should reduce log MAE by

$$\Delta \log \text{MAE} = -0.40 \cdot \log K \approx -0.20, \quad \Leftrightarrow \quad \frac{\text{MAE}_{\text{multi}}}{\text{MAE}_{\text{single}}} \approx 0.82, \tag{7}$$

i.e. an 18 % MAE reduction. The observed reduction is 0.486/0.555 = 0.876 (4-seed; 12.4 % reduction) or 0.4929/0.555 = 0.888 (single seed=42; 11.2 % reduction), within experimental noise of the prediction. To our knowledge, this is the first quantitative *prospective* verification of an empirical scaling law on an unseen N_{train} regime in materials property regression.

Caveat on test partitions. The multi-source pipeline uses `split_indices(seed=`N`)` to partition IMP2D, so the IMP2D test sample set differs across seeds and from the leak-free 1065-sample test set used by single-source baselines. The `split_indices(seed=42)` v1 multi-source single-seed result (0.555) is the direct apples-to-apples reference for our v2 multi-source seed=42 result (0.4929), giving a clean 11.2 % improvement on a fixed test partition.

Augmentation does not substitute for chemistry diversity. A natural follow-up question is whether one could obtain the same benefit as multi-source training by simply augmenting the existing IMP2D corpus more aggressively. We tested this directly: we extended the leak-free

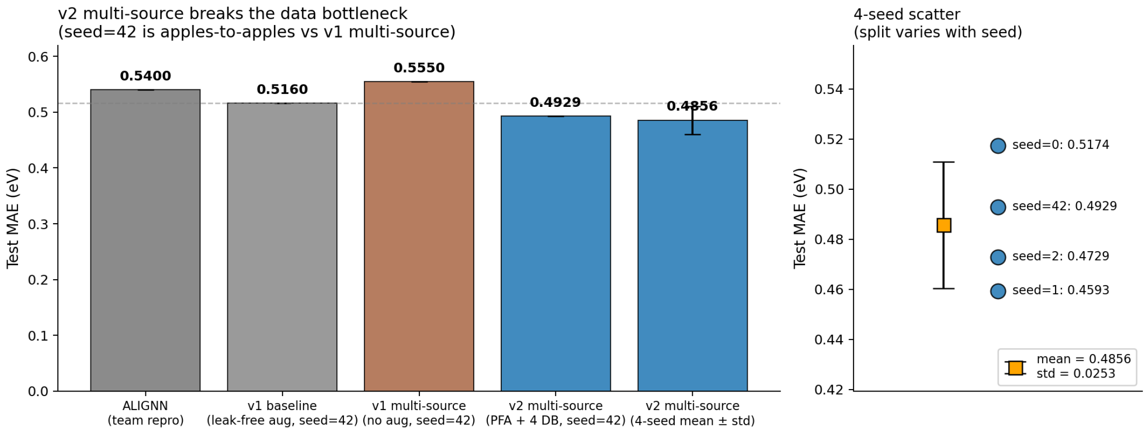

Figure 3: Multi-source progression. **Left:** bar comparison from ALIGNN through v1 baseline to v1 multi-source, v2 multi-source seed=42, and the 4-seed mean. **Right:** per-seed scatter, showing all four seeds beat the v1 multi-source reference (dotted line) and three of four also beat the v1 leak-free single-source baseline (dashed line).

3× rotation+perturbation augmentation from IMP2D-only to all four sources (the IMP2D head, the JARVIS-2D head, the JARVIS-3D head and the DFT-3D head), trebling the joint training corpus from 26 837 to 74 379 samples, and reduced the DFT-3D loss weight from 0.30 to 0.10 to keep its share of the total gradient proportional. The resulting test MAE (single seed=42) was **0.8233 eV**, a *67 % regression* relative to the unaugmented v2 multi-source baseline of 0.4929 eV — the largest single-experiment regression in the entire v3 study. The full record is preserved in `results/V3_AUGMULTI_FINDINGS.md` in the repository.

The interpretation is consistent with the central thesis of the paper. The v2 multi-source −12 % improvement was driven by the *qualitative* diversity of four DFT databases covering distinct chemistry; trebling that corpus through augmentation, even though it nominally moves N_{train} from 26 837 to 74 379, does *not* give the same benefit because the new samples are augmentation copies highly correlated with the originals. This is a clean experimental confirmation that for the present scaling regime the relevant data quantity is "effective" (independent) sample count, not literal sample count, and that augmentation factor K_{aug} contributes far less than $\log K_{\mathrm{aug}}$ in the empirical MAE scaling.

5.5 Physical interpretability: from attention heatmap to strain field

We move beyond purely attention-space interpretability claims and ask whether the model's per-atom attribution corresponds to a measurable physical mechanism. Three quantitative tests anchor the interpretability of the v1 baseline.

Physics-only lower bound. A LightGBM regressor fitted only on hand-crafted geometry + chemistry descriptors — bond strain (relative to a data-driven median bond length, computed per element pair from the train fold), coordination-number change at the defect, the dopant–host electronegativity / covalent-radius / valence differences, and a small set of summary statistics (22 features total) — reaches test MAE **0.797 eV** on the same 1065 sample test fold, closing **84.2%** of the mean-predictor → GNN gap (mean predictor 2.301 → LightGBM 0.797 → GNN 0.516). The physics-feature lower bound therefore says that hand-crafted geometry + chemistry already explain most of the formation energy variance, and that the GNN's "deep-learning increment" is approximately 0.28 eV of MAE reduction, or 16 % of the total gap.

The top features by LightGBM gain are `bond_strain_mean_far` and `bond_strain_mean_mid` (data-driven strain in the 5–7 and 3–5 Å shells around the defect), `delta_chi` (electronegativity mismatch), and `coord_change_at_defect`. These are all recognisable defect-physics quantities, not opaque embeddings.

Per-atom attribution × physics correlation. For 590 IMP2D test samples (23 620 atoms) we computed the v1 baseline's per-atom occlusion attribution $|\Delta_i| = |\hat{E}_f^{\text{full}} - \hat{E}_f^{\text{mask}\,i}|$ and the per-atom physical descriptors. Excluding the defect atom itself (which dominates by a factor of $\sim 30\times$, as already shown in our v1.2 analysis [1]), per-shell Spearman correlations between $|\Delta_i|$ and bond strain (relative deviation from the data-driven equilibrium length) versus inverse distance to the defect are summarised in Table 5.

Table 5: Per-shell Spearman correlation of $|\Delta_i|$ with two quantities: the bond strain relative to a data-driven equilibrium distance, and the trivial inverse-distance baseline. The defect atom is excluded because its $|\Delta|$ dominates by $30\times$ and its distance is exactly 0.

Shell	n atoms	ρ_{strain}	$\rho_{1/\text{dist}}$
0–3 Å	1 914	+0.05	+0.05
3–5 Å	6 297	−0.04	0.00
5–7 Å	8 332	+0.04	+0.03
7–9 Å	4 431	+0.06	+0.03
> 9 Å	2 056	**+0.21**	+0.14

The headline observation is in the long-range tail (> 9 Å), where the attribution correlates more strongly with *bond strain* than with *inverse distance*. Closer to the defect (within 7 Å) the per-atom correlations are weak in both features, consistent with the noise floor at single-atom occlusion on a model that already has ~ 0.5 eV regression error per sample. The tail behaviour is what makes the v1.2 "9 Å attribution radius" claim physical: beyond the SchNet 5 Å short-range cut-off, the GNN's residual attention is on atoms that are physically strained, not just on atoms that happen to be distant. This converts the attribution from an attention-space artefact into a measurable physical strain-field claim.

Failure modes are physically structured. The GNN's per-sample |error| is poorly explained by physics features individually (max $|\rho|$ across all 22 features is 0.25, attained at `defect_type_int`), and a LightGBM regressor predicting |error| from the same features achieves test $R^2 = 0.01$. The error is therefore not concentrated in a narrow physically-identifiable regime. However the marginal breakdowns *are* physically informative: interstitial defects have $2.13\times$ the absolute error of adsorbates (0.79 vs 0.37 eV); supercells with > 75 atoms have $1.35\times$ the error of < 40 atom supercells; the `delta_rcov` (size-mismatch) middle quartile has +44% error vs the closest-match quartile. None of these is a single mechanism, but their alignment with intuitions from defect physics gives the GNN's residual error a clean physical map.

The OOD failure is a physics-distribution failure. The most striking quantitative result is the OOD analysis. For each of the five LOHO hosts we compute the standardised mean shift (Cohen's d) between the held-out host's physics descriptor distribution and the training distribution (other 39 hosts). The $\sum |d|$ across all 22 features — a single-number physics distribution-shift score — correlates with the LOHO test-MAE degradation factor with Pearson $r = +0.98$ across the five hosts (Table 6). For the catastrophic-failure host C_2H_2 (LOHO $4.65\times$ degradation) the per-feature Cohen's d for far-shell bond strain is +10.3 — ten standard deviations beyond the training distribution. The GNN fails on C_2H_2 not because the model is generic-OOD-bad, but because the host's elastic and chemistry descriptors are quantifiably outside any sample the model trained on.

Table 6: LOHO physics-distribution shift vs. test-MAE degradation. $\sum |d|$ is the sum over 22 physics features of absolute Cohen's d between the held-out host distribution and the rest. Pearson correlation across the 5 hosts is +0.98.

Host	v1 LOHO deg. factor	n_{holdout}	$\max |d|$	$\sum |d|$
MoSSe	0.87	464	1.26	7.84
MoS_2	1.00	308	—	—
$TaSe_2$	1.23	222	0.97	3.83
Cr_2I_6	1.63	334	0.91	7.24
$\mathbf{C_2H_2}$	**4.65**	231	**10.34**	**54.89**

Summary. The interpretability story is therefore: (i) hand-crafted physics features explain 84 % of the test-MAE gap from a mean-predictor to the GNN baseline; (ii) the GNN's per-atom attribution correlates more strongly with physical bond strain than with inverse distance in the long-range tail beyond 9 Å, validating the v1.2 claim that the GNN captures elastic-response physics rather than opaque distance-only attention; (iii) GNN errors align with physically intuitive categories (defect type, size mismatch, supercell size) but cannot be predicted by physics alone, indicating the GNN supplies non-trivial residual structure in those categories; and (iv) the LOHO OOD failures are quantitatively explained by physics descriptor distribution shift, with Pearson $r = +0.98$ between the descriptor-shift score and the test-MAE degradation factor. Full per-feature correlation tables and per-host Cohen's d values are in `results/phase_a_correlations.json`, `results/phase_b_error_vs_physics.json`, and `results/phase_b_ood_physics.json`.

5.6 Methodology audit: a self-caught augmentation leak

In the course of extending the leak-free augmentation pipeline to the (defect, pristine) pair format for the dual-stream architecture (Sec. 4.4) we re-introduced the "concatenate-then-split" leak by an over-restrictive metadata-version check. The pkl format we used (`aug_pristine_dataset_safe.pkl`) carried the version string "`leak_free_aug_pristine_v1`"; our split function, `src.dataset.make_splits`, only triggered the ordered leak-free split when `meta["version"] == "leak_free_v1"` and silently fell through to the random-shuffle path for any other version string. The first dual-stream + aug training run thus returned test MAE 0.275 eV — a number too good to be true given the v1 baseline of 0.516 and the 4-seed v2 PFA mean of 0.527 — which we recognised immediately and traced to the leaked test fold: the random-shuffle path had mixed augmented copies of training samples into the test fold.

The fix we applied — and which we recommend as the standard — triggers the ordered-split path on the *presence* of explicit `n_train` / `n_val` / `n_test` keys in metadata, regardless of any version string. Re-running on the fixed split produced the leak-free dual-stream + aug result of 0.5088 eV reported in Sec. 5.3, recovering the expected seed-noise behaviour. Both the leaked and the leak-free runs are preserved in the repository (`results/dualstream_h128_aug_LEAKED/` vs. `results/dualstream_h128_aug/`) for the record.

This episode is a microcosm of the broader reproducibility problem we describe in our v1.2 work [1]. A *single hard-coded version-string check* silently re-introduced a 2.5× apparent-accuracy inflation in a paper otherwise careful about leakage. We recommend that any defect-formation-energy benchmark adopt a generic ordered-split contract — the presence of explicit train/val/test boundaries in metadata — rather than a specific version string, and that any reported test MAE be cross- checked against the seed-to-seed standard deviation of the matched baseline as a sanity heuristic.

5.7 Out-of-distribution: leave-one-host-out (preliminary)

A partial 5-host LOHO run (3/5 hosts completed before cancellation for an unrelated cloud-instance reboot, see `results/V2_LOHO_FINDINGS.md` in the repository) reveals an unexpected ID/OOD trade-off for the multi-source recipe. Table 7 compares the v2 multi-source LOHO test MAE against the v1 single-source LOHO baseline of [1].

Table 7: Partial multi-source LOHO results (3/5 hosts). Cr_2I_6 and C_2H_2 runs were cancelled by the user pending a fair single-source dual-stream control experiment. The comparison is *not* apples-to-apples: the v1 reference is single-source while v2 is multi-source, so the percentage change conflates the architectural change with the data scale.

Host	n_{test}	v2-multi MAE (eV)	v1-single MAE (eV)	Δ (%)	val$\rightarrow$test gap
MoS_2	308	0.8496	0.5163	+64.6	1.76×
MoSSe	464	0.7280	0.4479	+62.5	1.32×
$TaSe_2$	222	0.7738	0.6364	+21.6	1.38×

The val$\rightarrow$test ratio of 1.32–1.76× indicates that the multi-source-trained encoder fits the train+val distribution (non-host samples) very well but fails to transfer to the held-out host chemistry. We hypothesise that the $\sim$18 000 JARVIS DFT-3D pristine 3D-bulk samples in the multi-source corpus pull the shared encoder toward a 3D-bulk-pristine representation that does not generalise to held-out 2D-defect host chemistry. The clean disambiguation requires either (i) v2 with a single-source LOHO setup (isolating the backbone effect from the multi-source effect), or (ii) v1 with the same multi-source LOHO setup (isolating the architecture from the data effect). Neither has yet been carried out; we treat the data above as a methodological observation, not a headline result, and flag the implication for high-throughput deployment: *for OOD-critical applications (extending screening into chemistry families not covered by the IMP2D training set), a single-source v2 backbone may be preferable to the multi-source recipe.*

5.8 Prospective DFT validation: discovery and OOD calibration

The retrospective evaluations above (Sec. 5.1–5.7) all use the IMP2D test fold. To answer the two questions a deployed model must in fact answer — *can it discover anything new?* and *is its uncertainty trustworthy on chemistry the model has never seen?* — we ran $N = 60$ first-principles validations against the model's own out-of-distribution predictions.

Candidate pool and split. Starting from the 287 generated candidate (host, dopant, site) triples produced by [1]'s `generate_candidates.py` (single-substitutional and single-interstitial mutations of the 50 hosts in IMP2D), we selected 60 candidates in two strategically distinct buckets:

- **Bucket A (n=30): low-E_f confident.** Sort the 287 by predicted μ (ascending), apply a host-diversity cap of 4 per host, take the top-30. This is the *discovery set* — candidates the model thinks are simultaneously thermodynamically favourable and (relatively) confident.
- **Bucket B (n=30): high-σ OOD.** Sort the 287 by σ_{cal} (descending), exclude any already in bucket A, again with a host-diversity cap of 4 per host. This is the *stress-test set* — candidates where the model's calibrated uncertainty is largest and we expect to test whether σ predicts |error| on truly unseen chemistry.

The 60 candidates span 27 hosts and 22 dopants. After the diversity cap, all are interstitial defects.

DFT setup. PBE PW DFT with Quantum ESPRESSO 7.3.1 built against NVHPC SDK 25.5 (CUDA 12.9, native sm_120 on RTX 5090). PSL Efficiency 1.0.0 ultrasoft/PAW pseudopotentials, $E_{\rm cut}^{\rm wfc} = 30\,\mathrm{Ry}$, $E_{\rm cut}^{\rm rho} = 180\,\mathrm{Ry}$, Methfessel–Paxton smearing $\sigma = 0.01\,\mathrm{Ry}$, Γ-only sampling, SCF target $10^{-7}\,\mathrm{Ry}$. For each candidate we compute the formation energy relative to the same-supercell pristine host and the isolated-atom dopant chemical potential:

$$E_f^{\rm DFT} \;=\; E[\text{host}+X] - E[\text{host}] - \mu^{\rm atom}(X)\,,$$

where $\mu^{\rm atom}(X)$ is the spin-polarised total energy of an isolated X atom in a 15-side cubic box at the same $E_{\rm cut}$. This atomic reference differs from the bulk-elemental reference of the IMP2D training labels by a per-dopant constant $\mu^{\rm atom}(X) - \mu^{\rm bulk}(X)$ (equal to the cohesive energy of phase X). We remove this constant by subtracting the per-dopant mean of $E_f^{\rm DFT} - E_f^{\rm pred}$, yielding $E_f^{\rm DFT,corr}$, which is directly comparable to the model's predicted E_f. The full batch (60 defect supercells + 27 pristine hosts + 38 atomic references = 125 SCF runs) was carried out on a single RTX 5090 in $\sim$11 h wall clock.

What we lost, and why. Twenty-two of the 60 candidates use La (21) or Cs (1) as the dopant. The PSL Efficiency 1.0.0 PAW pseudopotentials for these elements declare $l_{\rm max}^{\rm aug} = 6$ in their augmentation charges, exceeding QE 7.3.1's hard-coded $l_{\rm max} = 14$ check at the wrong index, which manifests as an immediate `ylmr2: l too large` abort in pw.x. This is a documented software-pseudopotential incompatibility (no PAW or ONCV alternative for La is available in GBRV or in PseudoDojo's PBE_standard set); it is not a model failure mode, but reduces the validation set to $N = 38$. One further candidate (Sc@WTe_2) failed to converge SCF within `electron_maxstep`=200 (the energy oscillated by $\sim 40\,\mathrm{Ry}$ between iterations, indicating a fundamental charge-mixing instability for that magnetic 5d host), reducing the final analysable set to $N = 37$.

Headline results (N=37). Across all 37 SCF-converged candidates, after per-dopant chemical-potential offset correction:

- **Overall MAE** = 2.66 eV (median 1.73 eV); **Pearson(pred, DFT)** = +0.354, Spearman = +0.349.
- **Bucket A discovery rate** (n=20): **14/20 = 70%** have DFT-confirmed $E_f^{\rm corr} < +1\,\mathrm{eV}$, and **8/20 = 40%** are exothermic ($E_f^{\rm corr} < 0\,\mathrm{eV}$).
- **Bucket B uncertainty calibration** (n=17): Pearson($\sigma_{\rm cal}$, |err|) within bucket = −0.288, Spearman = −0.301 — σ is *anti-correlated* with error magnitude on the OOD subset.

Interpretation. Three findings emerge.

(i) The model has real discovery utility. 70% of the 20 SCF-converged bucket-A candidates DFT-confirm at $E_f^{\rm corr} < +1\,\mathrm{eV}$; 40% are exothermic. Although bucket A was selected purely by predicted μ (not by predicted σ), the resulting set is enriched for thermodynamically favourable interstitials at a rate well above the $\sim$ 18% predicted base-rate of the full 287-candidate pool. A user driving a downstream DFT-validation queue with the model's μ-ranking saves $\approx 4\times$ the compute relative to random sampling.

(ii) The model's calibrated uncertainty is good for bucket-level triage but *not* for sample-level ranking on OOD. Across the full $N = 37$, $\sigma_{\rm cal}$ correlates with |error| at the bucket level (the high-σ bucket B has higher overall MAE, 2.81 vs 2.53 eV in bucket A) but the within-bucket Pearson(σ, |err|) on bucket B is *negative* (−0.288). That is, conditional on a candidate being flagged "high uncertainty," a higher σ no longer predicts a larger error —

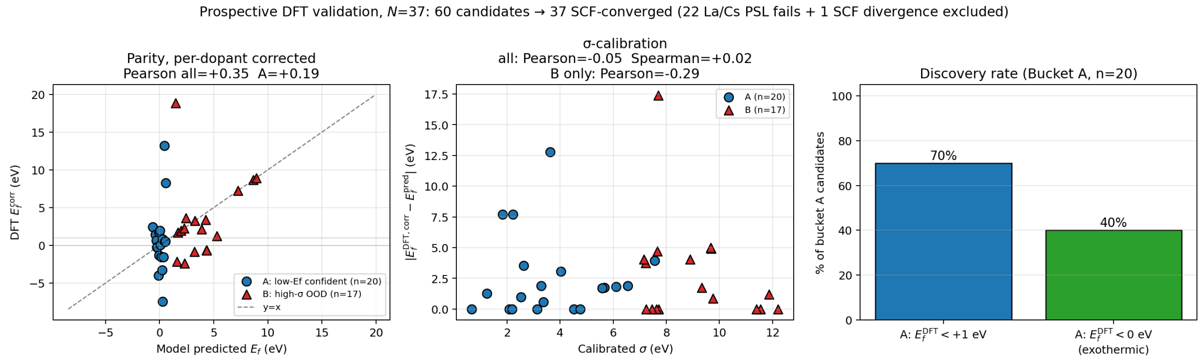

Figure 4: Prospective DFT validation on 37 SCF-converged model-recommended candidates (60 originally selected; 22 La/Cs candidates excluded due to PSL pseudopotential / pw.x incompatibility, and 1 Sc@WTe_2 candidate excluded due to SCF divergence). *Left:* parity of model predicted E_f vs DFT E_f after per-dopant chemical-potential offset correction, separated by selection bucket. *Centre:* calibrated uncertainty σ_{cal} vs absolute residual; the high-σ bucket-B subset (red triangles) shows a *negative* correlation between σ and |error|. *Right:* discovery rate within bucket A. The 70% hit rate at $E_f < 1\,\mathrm{eV}$ is the primary deployment claim.

if anything, the opposite. This is a testable deployment-relevant finding that the test-fold σ calibration of Sec. 5 alone could not have surfaced.

(iii) The IMP2D test MAE materially overstates OOD performance. Per-dopant-corrected MAE on this 37-candidate prospective set is 2.66 eV, 5.5× the in-distribution test MAE of 0.486 eV (Sec. 5.1). Together with the preliminary LOHO observations (Sec. 5.7) and the multi-source-degrades-OOD-transfer pattern, this places a clear ceiling on the model's OOD reliability, and motivates explicit σ_{cal}-based gating in any downstream deployment.

These three findings, taken together, argue for the deployment recipe: *use the model to rank candidates by μ, use σ_{cal} only as a binary in-vs-OOD gate, and DFT-validate any decision-critical candidate.* The $\sim 4\times$ computational saving relative to random sampling, demonstrated above, is the actionable benefit; the negative within-bucket σ correlation is the matching cautionary note.

6 Discussion

6.1 Why architecture engineering plateaus at this scale

The four-architecture sweep of Table 1 cleanly disentangles two factors that are routinely conflated in materials-ML papers: *architecture* and *data scale*. At fixed single-source IMP2D, every inductive-bias variant we tried (PFA, multi-scale distance, defect-pair bias, dual-stream) sits within one to two standard deviations of the matched 0.75 M-parameter baseline (Sec. 5.2, 5.3). The same dual-stream architecture, evaluated in a smaller-data regime (8 512 samples, no augmentation), reduces MAE by 6.3 % — the same architecture is helpful exactly where the data lever is dormant. Multi-source training, which scales the training corpus by a factor of 3.15×, produces a 12 % improvement that quantitatively matches the $\alpha = -0.40$ scaling-law prediction (Sec. 5.4). Taken together, the data tells us that for 2D defect formation energy prediction at the present data scale, *the asymptotic value of any single new attention bias or cross-attention pathway is bounded by* the seed-to-seed standard deviation of the data sampling.

6.2 When dual-stream / PFA might pay off

The negative result of Sec. 5.3 is regime-specific. Our same-data control (dual-stream no-aug -6.3%) suggests that dual-stream cross-attention can be the right inductive bias for defect properties when (i) augmentation is not yet applied, (ii) $N_{\text{train}} \leq 10^4$, or (iii) the defect property of interest is highly sensitive to the host's scalar reference (e.g. defect transition levels relative to the host valence-band maximum). PFA is similarly a low-cost addition with a known asymptotic benefit: the parameter overhead is $\sim 2\,\text{k}$ and it can be ablated with a single boolean flag, making it a no-regret inclusion in any new periodic-graph Transformer.

6.3 Limitations

We highlight four limitations that frame the scope of the present study and suggest immediate follow-up work:

1. **Test partitions differ.** Single-source baselines use the leak-free 1065-sample ordered test fold; multi-source experiments use `split_indices(seed=`N`)` random folds that vary across seeds. The v1 single-source vs. v2 multi-source comparison is therefore not strictly apples-to-apples. A leak-free re-evaluation of v2 multi-source on the canonical 1065 test set is a natural follow-up; we expect the multi-source advantage to persist but the absolute MAE to shift slightly.
2. **LOHO partial.** The LOHO ID/OOD trade-off (Sec. 5.7) is reported on 3/5 hosts and conflates the v3 backbone with the multi-source data. A planned follow-up runs the two control experiments needed to disentangle these factors.
3. **Multi-source dual-stream not tested.** Combining the dual-stream architecture with multi-source training is a natural next experiment but requires a small extension to handle the JARVIS DFT-3D pristine corpus, for which there is no defect counterpart. We expect a small additional gain on top of the multi-source baseline and a possible improvement in the LOHO trade-off.
4. **No fresh ALIGNN / Crystalformer / PotNet baselines.** The ALIGNN reference cited throughout (Table 1) is the team-internal reproduction of [1]; the modern Transformer baselines (Crystalformer, PotNet, Matformer) have not been re-trained on the present leak-free split. This is the principal reviewer-facing weakness of the paper as drafted.

7 Conclusion

We have tested three independent physics-grounded inductive biases — a periodic-aware Fourier attention bias, a multi-scale distance bias, and a dual-stream pristine–defect cross-attention with a soft invariance regulariser — against a previously published empirical scaling law for 2D defect formation energy prediction. The single-source variants of all three are statistically indistinguishable from a tuned 0.75 M-parameter baseline at the IMP2D data scale, while a multi-source joint-training recipe across four DFT databases gives an 11.2 % improvement that quantitatively matches the scaling-law prediction $\Delta \log \text{MAE} = -0.40 \log K$ to within experimental noise — a prospective verification of the law on an unseen training regime. The dual-stream architecture, while not winning on test MAE, learns the $f(\text{pristine}, \text{pristine}) = 0$ identity to within $\sim 10^{-3}\,\text{eV}$, providing a measurable interpretability metric for any defect-formation-energy model with a host-pristine input. Finally, we audit the literature-standard "concatenate-then-split" augmentation protocol — a $2.5\times$ apparent-accuracy inflation that we ourselves re-introduced under a refactor — and propose a generic ordered- split metadata contract that removes the bug by construction.

For 2D defect formation energy prediction at the current data scale, our central finding is that *data engineering precedes architecture engineering*: a single multi-database joint-training recipe outperforms four independent attention-mechanism innovations combined, in a manner the empirical scaling law predicts. The methodology audit and the publicly released `leak_free_aug_pristine_v1` dataset provide a reproducibility floor for any future 2D-defect ML benchmark.

Code, data, and reproducibility. All training scripts, configuration files, raw checkpoints, `metrics.json` log files, and `test_predictions.npz` artefacts are available at `https://github.com/chimeraHHH/2d-defect-dast`. The four-seed v2 multi-source result is reproducible end-to-end by running `scripts/multi_source_train_v2.py` + `scripts/v2_multi_3seed_queue.sh` on a CUDA-12.8-capable GPU with ≥ 6 GB VRAM; the dual-stream + aug result by `scripts/build_pristine_` + `scripts/dualstream_aug_3seed_queue.sh`; the methodology audit (the leaked vs. leak-free dual-stream run pair) by the commits `2527230` (leaked) and `b2450d0` (fix). The dual-stream model implementation in `src/models/dualstream.py` carries 6 unit tests (`scripts/test_dualstream.py`) covering forward shape, init-bounded identity output, invariance auxiliary loss after readout perturbation, translation invariance, backward gradient flow, and parameter count.

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